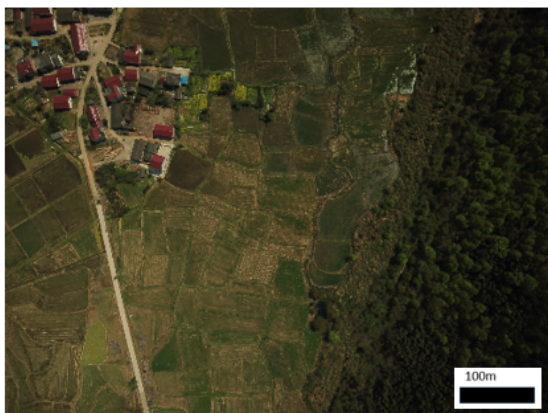


1 Appendix 1. Examples of fields in traditional (A) and consolidated farmland (B).

A.



B.



2

3

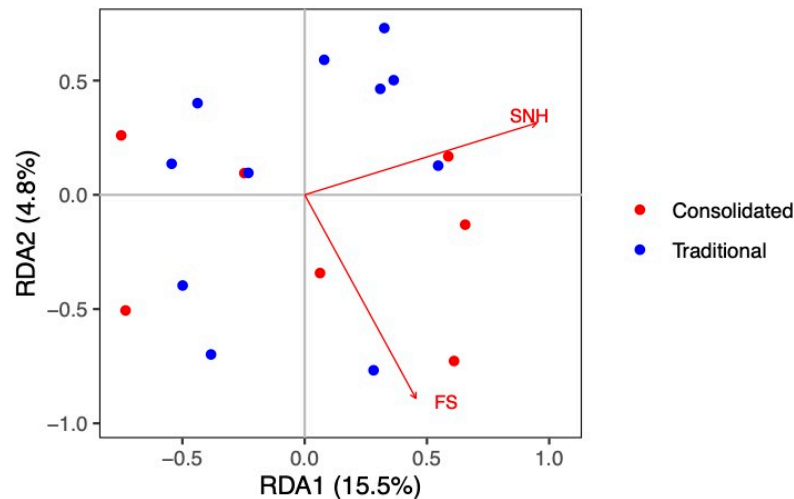
4 Appendix 2. Wild bee body size and abundances, with further division into above- and below-ground
5 nesting species.

6

Species name	Nesting location	Female body size (mm)	Consolidated farmland	Traditional farmland
<i>Andrena callopyrrha</i>	Underground	2.34	12	16
<i>Andrena delicatula</i>	Underground	2.57	0	2
<i>Andrena luridiloma</i>	Underground	1.95	19	49
<i>Andrena mediocalens</i>	Underground	2.21	0	2
<i>Andrena taraxaci</i>	Underground	1.68	1	0
<i>Andrena thoracica</i>	Underground	2.91	0	2
<i>Anthophora plumipes</i>	Underground	3.31	0	2
<i>Apis cerana</i>	Above ground	2.62	94	136
<i>Ceratina flavipes</i>	Above ground	1.78	12	37
<i>Ceratina japonica</i>	Above ground	1.71	0	2
<i>Ceratina okinawana</i>	Above ground	1.22	43	69
<i>Ceratina smaragdula</i>	Above ground	1.41	2	17
<i>Eucera floralia</i>	Underground	2.86	208	120
<i>Habropoda sinensis</i>	Underground	2.96	0	2
<i>Halictus aerarius</i>	Underground	1.03	1	18
<i>Halictus tsingtouensis</i>	Underground	2.13	13	24
<i>Lasioglossum halictoides</i>	Underground	2.67	107	68
<i>Lasioglossum occidens</i>	Underground	2.26	1	5
<i>Lasioglossum proximaturn</i>	Underground	2.43	127	155
<i>Lasioglossum scitulum</i>	Underground	2.31	70	51
<i>Lasioglossum sp1</i>	Underground	2	50	57
<i>Lasioglossum sp2</i>	Underground	2.42	60	84
<i>Lasioglossum sp3</i>	Underground	2.36	4	33
<i>Lasioglossum sp4</i>	Underground	2.15	3	3
<i>Lasioglossum sp5</i>	Underground	2.33	24	16
<i>Lasioglossum sp6</i>	Underground	2.32	29	0
<i>Lasioglossum subopacum</i>	Underground	2.89	97	132
<i>Osmia cornifrons</i>	Above ground	2.86	0	9
<i>Osmia excavata</i>	Above ground	2.73	1	16
<i>Osmia microdonta</i>	Underground	2.33	0	1
<i>Osmia pedicornis</i>	Above ground	2.88	3	12
<i>Osmia sp1</i>	Above ground	2.71	0	2
<i>Xylocopa rufipes</i>	Above ground	4.59	1	1
<i>Xylocopa tranquebarorum</i>	Above ground	4.5	5	7

7

Appendix 3. Redundancy analysis (RDA) of wild bee communities, constrained by the amount of semi-natural habitat (SNH) and field size (FS). RDA axis 1 and 2 explained 15.5% and 4.8% of the distribution, respectively. The adjusted R^2 was 0.08. The full model is significant ($F = 1.73$, $p = 0.021$, F distribution based on 999 permutations). Variation in pollinator community composition was captured by the first axis ($F = 2.65$, $P = 0.02$), while responses on the second axis were non-significant ($F = 0.8$, $P = 0.695$). Based on permutation test ($n = 999$), the impact of SNH was significant ($p = 0.01$), while the effect of FS was not ($p = 0.451$).



Appendix 4. The impact of farmland consolidation and the proportion of semi-natural habitat on the body size of *Lasioglossum proximatum* and *Eucera floralia*. Both full and final models include consolidation status (CS; base: traditional farmland), proportion of semi-natural habitat (SNH) and the interaction between semi-natural habitat and consolidation status (SNH*CS). None of the result was significant ($p < 0.05$).

	Model type	SNH	CS	SNH*CS
<i>Lasioglossum proximatum</i>	Full model	-0.02 ± 0.08	-0.03 ± 0.02 ($p = 0.06$)	0.08 ± 0.09
	Final model	/	-0.02 ± 0.01 ($p = 0.06$)	/
<i>Eucera floralia</i>	Full model	-0.08 ± 0.08	-0.02 ± 0.03	0.09 ± 0.09
	Final model	/	0.002 ± 0.01	/

27 Appendix 5. Non-metric multidimensional scaling (NMDS) plots based on Chord-Normalized Expected
28 Species Shared (CNESS) dissimilarity for $m=1$ (A, stress=0.14) and $m=20$ (B, stress=0.21) for wild pollinator
29 communities at the 18 research sites.

